Project Report: Quiz Application Using File Handling in C++

Project By:

Atulit Singh	17
Dhruav Mathur	27
Janshrutt Narula	20
Ashish Ohol	28

TABLE OF CONTENTS

Introduction
Problem Statement
List of OOP concepts
Explanation of concepts
Output Screenshots
Conclusion

Introduction

The objective of this project is to develop a Quiz Application that allows users to take quizzes, view their scores, and check a leaderboard. Additionally, the application provides administrative functionalities for managing quiz questions and viewing scores. The application uses file handling to store user information and quiz questions, ensuring data persistence across sessions. The implementation is done in C++ and leverages object-oriented programming (OOP) principles to enhance code organization and reusability.

Problem Statement

In educational settings, quizzes are a common method for assessing knowledge. However, managing quizzes, tracking user scores, and allowing users to register and log in can be cumbersome without a systematic approach. The project aims to address the following problems:

- **User Management**: Allow users to register, log in, and view their scores.
- **Quiz Management**: Enable the admin to add, delete, and manage quiz questions.
- **Score Tracking**: Maintain a leaderboard to track user scores.
- **Data Persistence**: Store user and quiz data in files to ensure that information is retained between sessions.

List of OOP Concepts Used in This Project

1. **Classes and Objects**:
    - The application defines multiple classes (**student**, **question**, and **admin**) to encapsulate related data and functionalities. Each class represents a specific entity in the application.
2. **Encapsulation**:
    - Data members of classes are made private or protected, and public methods are provided to access and modify these members. This helps in protecting the integrity of the data.
3. **Inheritance**:
    - The **question** class inherits from the **student** class, allowing it to access protected members and methods, promoting code reuse.
4. **Static Members**:

- Static members (e.g., **current_score** in the **student** class) are used to maintain shared data across all instances of a class. This is useful for tracking scores that are not tied to a specific user instance.

5. **Method Overloading**:
    - The application uses method overloading to provide multiple versions of methods (e.g., **setdata** and **setalldata** in the **student** class) that accept different parameters, enhancing flexibility.
6. **Data Abstraction**:
    - The implementation abstracts complex operations (e.g., file handling, score calculations) behind simple method calls, making the code easier to understand and use.

Explanation of Key Components

1. Student Class

- **Functionality**: Manages user registration, login, and score tracking.
- **Methods**:
    - **register_new_user()**: Registers a new user and stores their information in a file.
    - **login()**: Authenticates the user based on username and password.
    - **leaderboard()**: Displays the leaderboard by sorting users based on their scores.

2. Question Class

- **Functionality**: Handles quiz question management.
- **Methods**:
    - **get_new_question()**: Prompts the admin to enter a new question and its options.
    - **takequiz()**: Facilitates the quiz-taking process, including displaying questions and recording answers.
    - **getquestions()**: Reads questions from a file and returns them as a vector.

3. Admin Class

- **Functionality**: Provides administrative capabilities for managing quiz questions and viewing scores.

- **Methods**:

- **addquestions()**: Allows the admin to add new questions to the quiz.
- **removequestions()**: Enables the admin to delete a specific question from the quiz.
- **viewscores()**: Displays scores of all users.

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.22631.4317]
(c) Microsoft Corporation. All rights reserved.

D:\CODES-C++>a.exe

LOGIN AS:
1. Student
2. Admin
3. Exit
Your Choice: 1

1. Register
2. Login
Enter Your Choice: 2
ENTER USERNAME: aryan
ENTER PASSWORD: abc
LOGGED IN

1. TAKE QUIZ
2. VIEW SCORE
3. VIEW STUDENT DETAILS
4. VIEW LEADERBOARD
5. Exit
YOUR CHOICE: 1

What is the process by which water vapor turns into liquid water?
a) Evaporation
b) Condensation
c) Sublimation
d) Precipitation

YOUR ANSWER: b
What is the capital of France?
a) Berlin
b) Madrid
c) Paris
d) Rome

YOUR ANSWER: c
Who was the first President of the United States?
a) Abraham Lincoln
b) Thomas Jefferson
c) George Washington
d) John Adams

YOUR ANSWER: a
Who wrote the play 'Romeo and Juliet'?
a) Charles Dickens
b) William Shakespeare
```

```
C:\Windows\System32\cmd.e

b) William Shakespeare
c) Mark Twain
d) Leo Tolstoy

YOUR ANSWER: b
Which is the largest planet in our solar system?
a) Earth
b) Mars
c) Jupiter
d) Saturn

YOUR ANSWER: a
Which gas do plants absorb from the atmosphere during photosynthesis?
a) Oxygen
b) Carbon Dioxide
c) Nitrogen
d) Hydrogen

YOUR ANSWER: b
What is the currency of Japan?
a) Yen
b) Won
c) Dollar
d) Peso

YOUR ANSWER: a
Which element has the chemical symbol 'O'?
a) Gold
b) Oxygen
c) Osmium
d) Silver

YOUR ANSWER: a
What is the smallest continent by land area?
a) Europe
b) Australia
c) Antarctica
d) South America

YOUR ANSWER: b
Which ocean is the largest in the world?
a) Atlantic Ocean
b) Indian Ocean
c) Arctic Ocean
d) Pacific Ocean

YOUR ANSWER: d
YOUR SCORE IS: 7
```

```
C:\Windows\System32\cmd.e

YOUR SCORE IS: 7

1. TAKE QUIZ
2. VIEW SCORE
3. VIEW STUDENT DETAILS
4. VIEW LEADERBOARD
5. Exit
YOUR CHOICE: 3


USERNAME: aryan
NAME: aryan sharma

1. TAKE QUIZ
2. VIEW SCORE
3. VIEW STUDENT DETAILS
4. VIEW LEADERBOARD
5. Exit
YOUR CHOICE: 4

NAME: aditya thakur
USERNAME: aditya
SCORE: 7

NAME: aryan sharma
USERNAME: aryan
SCORE: 7

NAME: dhruav mathur
USERNAME: dhruav
SCORE: 6

NAME: atulit singh
USERNAME: atulit
SCORE: 5

NAME: ashish ohol
USERNAME: ashish
SCORE: 4


1. TAKE QUIZ
2. VIEW SCORE
3. VIEW STUDENT DETAILS
4. VIEW LEADERBOARD
5. Exit
YOUR CHOICE: 5
```

```
C:\Windows\System32\cmd.e

LOGIN AS:
1. Student
2. Admin
3. Exit
Your Choice: 2
ENTER USERNAME: dhruav
ENTER PASSWORD: abc123
LOGGED IN

1. ADD QUESTIONS
2. VIEW SCORES
3. VIEW LEADERBOARD
4: REMOVE QUESTIONS
5. Exit
YOUR CHOICE: 2

dhruav          dhruav mathur           6
ashish          ashish ohol             4
atulit          atulit singh            5
aditya          aditya thakur           7
anuj            anuj singh              3
aryan           aryan sharma            7

1. ADD QUESTIONS
2. VIEW SCORES
3. VIEW LEADERBOARD
4: REMOVE QUESTIONS
5. Exit
YOUR CHOICE: 3

NAME: aditya thakur
USERNAME: aditya
SCORE: 7

NAME: aryan sharma
USERNAME: aryan
SCORE: 7

NAME: dhruav mathur
USERNAME: dhruav
SCORE: 6

NAME: atulit singh
USERNAME: atulit
SCORE: 5

NAME: ashish ohol
USERNAME: ashish
```

```
C:\Windows\System32\cmd.e

USERNAME: dhruav
SCORE: 6

NAME: atulit singh
USERNAME: atulit
SCORE: 5

NAME: ashish ohol
USERNAME: ashish
SCORE: 4


1. ADD QUESTIONS
2. VIEW SCORES
3. VIEW LEADERBOARD
4: REMOVE QUESTIONS
5. Exit
YOUR CHOICE: 4

1. Which is the largest planet in our solar system?
2. What is the capital of France?
3. Who wrote the play 'Romeo and Juliet'?
4. Which element has the chemical symbol 'O'?
5. What is the smallest continent by land area?
6. Which gas do plants absorb from the atmosphere during photosynthesis?
7. Which ocean is the largest in the world?
8. What is the currency of Japan?
9. What is the process by which water vapor turns into liquid water?
10. Who was the first President of the United States?
ENTER QUESTION NUMBER TO DELETE: 8

1. ADD QUESTIONS
2. VIEW SCORES
3. VIEW LEADERBOARD
4: REMOVE QUESTIONS
5. Exit
YOUR CHOICE: 5


LOGIN AS:
1. Student
2. Admin
3. Exit
Your Choice: 3


Thank you
Exiting...
D:\CODES-C++>
```

Conclusion

The Quiz Application effectively demonstrates the use of object-oriented programming principles to create a functional and maintainable software solution. By utilizing file handling for data persistence, the application provides a seamless experience for both users and administrators. This project not only addresses the problem of managing quizzes but also serves as a practical example of applying OOP concepts in C++. Future enhancements could include adding more features, such as timed quizzes, user feedback, and a graphical user interface (GUI).